

companies to update and change their infrastructure on a minute-by-minute basis, static dashboards and siloed applications were no longer sufficient in supporting the type of rapid-response environment created by cloud computing.

Traditional monitoring approaches struggled to keep pace for several reasons:

- They were built around fixed infrastructure assumptions.
- They lacked flexibility to model dynamic dependencies.
- They provided limited context across systems and teams.

The open source ecosystem started to fill these gaps by providing flexible, extensible, and standards-based alternatives that could evolve along with the infrastructure. It was also the beginning of a larger shift away from defining monitoring systems, towards behavioural observation of functionality.

Observability: The new baseline

Observability introduced a fundamentally different way of thinking about operations. Teams no longer asked pre-defined questions via static metrics; instead, they used their systems to discover failure types not yet known using real-time instrumentation and logging of data.

To support this new approach, three types of data became the basis of the observability model:

- **Metrics** for understanding performance patterns or trends related to service continuity.
- **Logs** for recording detailed information on events or actions that have occurred within the system.
- **Traces** for monitoring transaction processing across multiple components that exist across various environments or throughout networks.

The role of the open source community in the successful operationalisation of the observability model is very important for many reasons, including the widespread usage of Prometheus, Grafana, and OpenTelemetry tools. Not only did these tools succeed in being widely adopted as open source projects, but they also successfully filled gaps within many processes by providing visibility into operational problems and allowing for a shared standard of collecting telemetry across all platforms, environments, or programming languages.

As newer technology became available and companies continued to include visibility into their systems as a standard, observability also became popular as the new standard among modern IT operators.

Kubernetes: Creating a new control layer for managing resources and applications

When discussing modern-day operations, one cannot avoid discussing Kubernetes. Initially launched as a way of controlling the provisioning and orchestration of containers, Kubernetes has evolved to become an operational control layer for both applications and infrastructure.

Kubernetes simplified some areas of deployment and scaling; however, it created additional challenges through:

- Workloads being highly ephemeral.
- Service topologies changing quickly.
- Increased abstraction away from the core infrastructure.

Traditional methods of monitoring were rendered ineffective due to these dynamics; so open source tools were developed again to fill this void. With these Kubernetes-based observability, logging, and automation tools, organisations could gain visibility into their clusters, workloads, and the interactions between services.

Open source is now integral to operations related to Kubernetes. Most organisations find that usage of non-open source tools is not feasible for managing container-based environments.

Automation moves from convenience to a necessity

Automation has now matured to become a central function of how we operate our businesses. It started as a convenience, helping teams save time on routine tasks and minimising their manual effort in doing so, but when implemented on a larger scale, automation becomes critical for providing reliability.

Open source non-development automation made possible:

- Declarative provisioning of infrastructure.
- Consistency among systems environments using configuration management tools.
- Automated remediation of known failure scenarios.

Utilising IaC mitigated the potential for drift in the production environment and improved the repeatability of what is created in each cycle, while automation solutions enabled the automation of policies and standards without requiring ongoing human supervision.

The most effective operational models tightly integrated automation with observability — metrics and logs can trigger automated responses for incidents.

Adoption of open source in Windows and mixed OS enterprises

The increased usage of open source solutions within historically Windows-centric companies is one of the largest trends and shifts in this decade. Open source solutions were once seen as either incompatible with or peripheral to Windows environment operations but, today, open source components are embedded in almost every aspect of business activity conducted daily.

The trend towards using open source solutions in what were historically Windows-centric companies has occurred for several reasons including:

- A heterogeneous environment comprised of a mixture of both Windows and non-Windows operating systems (such as Linux) and cloud computing platforms.